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ROUND-TRIP TIMES

- The expected **round trip rate/restart** rate τ almost surely and in expectation equals,

$$\tau = \lim_{t \to \infty} \frac{R(t)}{t} = \lim_{t \to \infty} \frac{T(t)}{t} = \frac{N+1}{\mathbb{E}[\Delta T]}$$

- Under the ELE, we have round trip for each replica $R^m(t)$ defines a renewal process

$$\lim_{t\to\infty} \frac{T^m(t)}{t} = \lim_{t\to\infty} \frac{R^m(t)}{t} = \frac{1}{\mathbb{E}[\Delta T]}$$

- The time between round trips $\Delta T_i^m = T_{i,-}^m - T_{i-1,-}^m$ is i.i.d. in i and m

- By the key renewal theorem the following hold almost surely and in expectation:

- By conditioning on (I_1, ϵ_1) we obtain a recursive relationship for $f(i, \epsilon)$ which we can solve

- We can explicately compute the expected time between round times $\mathbb{E}[\Delta T]$

$$f(i, \epsilon) = \mathbb{E}[\Delta T \,|\, (I_0, \epsilon_0) = (i, \epsilon)]$$

- For $i \in \{0, \ldots, N\}$ and $\epsilon \in \{-1, 1\}$ define:

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ROUND-TRIP RATE FOR RPT